

APPENDIX C

C.1) RANDOM PULSES

(a) For a 5 second counting interval, $\langle N \rangle = 5$. Then

$$P(10 \text{ in } 5) = 5^{10} \exp(-5)/10! = 0.018 \quad .$$

(b) For a 1 second counting interval, $\langle N \rangle = 1$. Then

$$P(2 \text{ in } 1) = 1^2 \exp(-1)/2! = 0.18 \quad .$$

(c) $P(0 \text{ in } 5) = 5^0 \exp(-5)/0! = \exp(-5) = 0.0067 \quad .$

C.2) APPROACH TO GAUSSIAN DISTRIBUTION

We take the logarithm of (10) and insert the Stirling approximation (A.12b) for $\log N!$, dropping those terms that decrease with increasing N :

$$\begin{aligned} \log P(N) &= N \log \langle N \rangle - \langle N \rangle - N \log N + N - \frac{1}{2} \log(2\pi N) \\ &= -N \log(N/\langle N \rangle) + (N - \langle N \rangle) - \frac{1}{2} \log(2\pi N) \quad . \quad (S1) \end{aligned}$$

For large N , $\log N$ varies extremely slowly with varying N . So long as $\Delta N = N - \langle N \rangle$ remains small compared to $\langle N \rangle$, we may therefore approximate the second logarithm by replacing N by $\langle N \rangle$. This is not allowed, though, for the first logarithm, because its variation is magnified by the preceding factor N . We therefore retain N there. Replacing N by $\langle N \rangle$ in the last term, and setting $N = \langle N \rangle + \Delta N$ everywhere else:

$$\log P(N) = -(\langle N \rangle + \Delta N) \log(1 + \Delta N / \langle N \rangle) + \Delta N - \frac{1}{2} \log(2\pi \langle N \rangle). \quad (S2)$$

If now $\Delta N \ll \langle N \rangle$, the logarithms may be expanded as in (A.20):

$$\log(1 + \Delta N / \langle N \rangle) = \Delta N / \langle N \rangle - \frac{1}{2} (\Delta N)^2 / \langle N \rangle^2 + O(\Delta N^2) \quad .$$

If we insert this for the first logarithm in (S2), the result can be re-arranged to read

$$\log P(N) = - \frac{1}{2}(\Delta N)^2 / \langle N \rangle + O(\Delta N^3) - \frac{1}{2} \log(2\pi \langle N \rangle) \quad , (S3)$$

$$P(N) = (2\pi \langle N \rangle)^{\frac{1}{2}} \exp[-\frac{1}{2}(\Delta N)^2 / \langle N \rangle] \quad . \quad (S4)$$

This is of the form specified in the problem statement, with $A = (2\pi \langle N \rangle)^{\frac{1}{2}}$ and $B = \frac{1}{2} \langle N \rangle$.

Comment. If we had not replaced N by $\langle N \rangle$ in the last term in (S1), we would have obtained these additional terms in (S3):

$$- \frac{1}{2} \log(N / \langle N \rangle) = - \frac{1}{2} \log(1 + \Delta N / \langle N \rangle)$$

$$\cong - \frac{1}{2} \Delta N / \langle N \rangle + (\Delta N)^2 / 4 \langle N \rangle^2 \quad .$$

This changes the exponent in (S4) to

$$- \frac{1}{2 \langle N \rangle} (\Delta N)^2 \left(1 + \frac{1}{2 \langle N \rangle}\right) + \Delta N \quad \cong - \frac{1}{2 \langle N \rangle} [(\Delta N)^2 + \Delta N]$$

$$= - \frac{1}{2 \langle N \rangle} [(\Delta N + \frac{1}{2})^2 - 1/4] \quad .$$

The effect is a change of the probability maximum from $N = \langle N \rangle$ to $N = \langle N \rangle - \frac{1}{2}$, and a corresponding increase in the magnitude of the maximum probability by a factor $\exp(1/8 \langle N \rangle) \cong 1 + 1/8 \langle N \rangle$. For $\langle N \rangle \gg 1$ both changes are negligible.